

**JERUSALEM COLLEGE OF ENGINEERING**  
**(AN AUTONOMOUS INSTITUTION)**

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai  
Accredited by NBA, New Delhi and Accredited by NAAC with 'A' Grade  
Velachery Main Road, Narayanapuram, Pallikaranai, Chennai - 600100

|                      |                                                  |
|----------------------|--------------------------------------------------|
| Regulation: 2023     | Year/Sem/Branch: III / V / AI & ML               |
| Course Code: JAL2007 | Course Name: STORAGE TECHNOLOGIES                |
| Unit No: II          | Unit Title: INTELLIGENT STORAGE SYSTEMS AND RAID |

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

**UNIVERSITY EXAMINATION - MODEL ANSWER KEY (NOVEMBER 2026)**

**PART – B (16-MARK SYSTEM WORKFLOWS)**

**Question No. 3: Explain RAID technology in detail. Discuss RAID 0, RAID 1, RAID 3, RAID 5, and RAID 6 with their architecture, working principles, advantages, and limitations. (16 Marks)**

**EXAMINATION RESPONSE ABSTRACT:**

This document presents a comprehensive, university-grade examination response analysing Redundant Array of Independent Disks (RAID) technologies within enterprise storage systems. The response thoroughly decomposes the architectural models, underlying working principles, data protection mechanics, and performance characteristics associated with RAID Levels 0, 1, 3, 5, and 6. This analysis is designed to fulfil the learning objectives and cognitive levels prescribed in the Storage Technologies syllabus, bridging foundational concepts with practical enterprise deployment recommendations.

**Prepared By:**

Course Faculty

**Verified By:**

Course Coordinator

**Approved By:**

Head of Department (AI & ML)

# TABLE OF CONTENTS

|                                                                             |        |
|-----------------------------------------------------------------------------|--------|
| 1. Introduction to RAID Storage Technology                                  | Page 2 |
| 1.1 Limitations of Single Hard Disk Drives                                  | Page 2 |
| 1.2 Hardware RAID vs. Software RAID Implementations                         | Page 2 |
| 2. Core RAID Concepts & Parity Mathematics                                  | Page 3 |
| 2.1 Key Storage Techniques: Striping, Mirroring, and Parity                 | Page 3 |
| 2.2 Step-by-Step XOR Parity Reconstruction Walkthrough                      | Page 3 |
| 3. Detailed Deconstruction of RAID Levels: RAID 0, RAID 1, and RAID 3       | Page 4 |
| 4. Detailed Deconstruction of RAID Levels: RAID 5, RAID 6, and RAID 10      | Page 5 |
| 5. Comparative Analysis, RAID Write Penalty, and Enterprise Recommendations | Page 6 |

## 1. INTRODUCTION TO RAID STORAGE TECHNOLOGY

Redundant Array of Independent Disks (RAID) is an advanced storage virtualization technology that combines multiple physical hard disk drives (HDDs) or solid-state drives (SSDs) into a single logical storage unit. This logical grouping is presented to the operating system as a unified disk, allowing administrators to optimize performance, capacity, and data redundancy according to organizational requirements.

### 1.1 Limitations of Single Hard Disk Drives

Enterprise datacenters require continuous access to massive datasets. Relying on individual physical disks introduces severe bottlenecks and risks that make them unfeasible for modern operations. A single disk has critical limitations: (1) **Limited Storage Capacity:** restricted to its physical platter limits, unable to scale dynamically; (2) **Single Point of Failure (SPOF):** mechanical or electrical failure immediately makes the entire dataset unavailable, stopping business operations; (3) **Performance Bottleneck:** throughput is restricted by physical mechanisms, such as spindle rotational speeds (typically 7,200 to 15,000 RPM) and seek latency; (4) **Difficult Recovery:** recovering data from a failed disk has no native fallback mechanism.

### 1.2 Hardware RAID vs. Software RAID Implementations

| Architectural Feature | Hardware-Based RAID Controller                                                                                                                | Software-Based RAID Implementation                                                                                              |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Processing Overhead   | Offloaded entirely to a dedicated RAID processor on a separate RAID controller card, preserving the host server's primary CPU resources.      | Processes RAID calculations directly on the host server's primary CPU, increasing system overhead.                              |
| Write Cache & Backup  | Contains high-speed cache memory supported by non-volatile batteries to prevent cache loss, ensuring write data is safely committed to disks. | Relies on the host OS's volatile memory for caching, with no dedicated hardware backup, risking data loss during power outages. |
| Cost & Compatibility  | Requires PCIe card investment, but has high compatibility with diverse operating systems.                                                     | Cost-effective with integrated operating systems, but may have compatibility issues with older OS versions.                     |
| Performance Profile   | Outstanding performance for write-heavy calculational workloads, especially in high-density server environments.                              | Mixed performance; excels in read-heavy scenarios but may lag in write-intensive tasks.                                         |

## 2. CORE RAID CONCEPTS & PARITY MATHEMATICS

To address the performance and availability limitations of individual disk drives, RAID technologies leverage three primary foundational techniques: Disk Striping, Data Mirroring, and Parity.

### 2.1 Foundational Techniques Explained

- **Disk Striping (Performance):** Divides continuous logical data streams into small, uniformly sized segments (blocks, sectors, or bytes) and writes them sequentially across multiple disks. This enables parallel read and write requests across multiple active channels. It drastically improves throughput but provides zero fault tolerance.
- **Data Mirroring (Redundancy):** Stores identical copies of data on two or more distinct physical disks. When data is written, the controller duplicates the identical blocks across drives, providing high reliability. However, it incurs a substantial 50% storage overhead cost.
- **Parity (Cost-Effective Protection):** Parity is an extra mathematical correction block used to reconstruct missing data blocks. Instead of storing massive duplicate copies of data, RAID calculates parity values of active blocks using the Exclusive OR (XOR) binary logical operation. This provides full data protection with low overhead.

### 2.2 Mathematical Basis of Parity: Step-by-Step XOR Calculations

The mathematical operation driving parity in RAID levels (specifically RAID 3, RAID 5, and RAID 6) is the bitwise Exclusive OR (XOR) logic gate.  $P$  is defined as:  $P = A \oplus B \oplus C$ . If any single disk fails, say Disk B, its data can be reconstructed on a hot-spare disk by performing the XOR operation across the surviving data disks and the parity disk:  $B = A \oplus C \oplus P$ .

#### ACADEMIC EXAM STEP-BY-STEP WORKFLOW — DRIVE REBUILDING ANALYSIS:

Let us model a three-disk storage system where Disk 1 and Disk 2 hold user data, and Disk 3 stores the calculated parity. Assume an active 8-bit binary data block is being processed:

• **Step 1: Define Original Data Blocks:** Block A (Disk 1) = **10101100** | Block B (Disk 2) = **11001010**

• **Step 2: Calculate Parity P (to be stored on Disk 3):**

$P = A \oplus B = 10101100 \oplus 11001010 = \mathbf{01100110}$  (Calculated parity block)

• **Step 3: Simulate Physical Disk Failure & Recovery:**

Assume Disk 2 (holding Block B) fails. The system reconstructs Block B using Block A and Parity P:

$B = A \oplus P = 10101100 \oplus 01100110 = \mathbf{11001010}$

Observe that the reconstructed block is **11001010**, which perfectly matches the original Block B! This calculation mathematically proves that parity-based RAID can fully reconstruct lost information.

## 3. DETAILED DECONSTRUCTION OF RAID LEVELS: RAID 0, RAID 1, & RAID 3

### 3.1 RAID 0: Block-Level Disk Striping

RAID 0 divides continuous logical files into blocks and spreads them sequentially across multiple physical disks.

```
RAID 0 (STRIPING):  
Disk 1: [ Block A1 ] [ Block A3 ] [ Block A5 ]  
Disk 2: [ Block A2 ] [ Block A4 ] [ Block A6 ]
```

- **Working Principle:** Executing concurrent I/O requests across independent physical drive controllers maximizes bandwidth.

- **Advantages & Limitations:** Fastest performance (read/write) and 100% capacity utilization. However, it provides zero fault tolerance; a single drive failure destroys the entire array.

### 3.2 RAID 1: Disk Mirroring

RAID 1 writes identical data blocks to two distinct physical disks concurrently, creating a continuous copy.

```
RAID 1 (MIRRORING):  
Disk 1 (Primary): [ Block A1 ] [ Block A2 ] [ Block A3 ]  
Disk 2 (Mirror) : [ Block A1 ] [ Block A2 ] [ Block A3 ]
```

- **Working Principle:** Writes are duplicated. Reads can be split and processed in parallel across both disks, enhancing random read performance.

- **Advantages & Limitations:** Outstanding data safety and easy rebuilding. However, it has high storage costs, as usable storage efficiency is limited to exactly 50%.

### 3.3 RAID 3: Byte-Level Striping with Dedicated Parity

RAID 3 stripes data at byte granularity evenly across data disks, utilizing a single disk solely for calculated parity.

```
RAID 3 (BYTE-LEVEL STRIPING):  
Data Disks (1-3) : [ Byte A1 ] [ Byte A2 ] [ Byte A3 ]  
Parity Disk (4)  : [ Parity AP = Byte A1 ^ A2 ^ A3 ]
```

- **Working Principle:** Byte-level striping requires all drive spindles to rotate in perfect synchronization. Every I/O request accesses every physical disk in the array.

- **Advantages & Limitations:** Excellent sequential throughput (ideal for video streaming and large files). However, the dedicated parity disk creates a write bottleneck since it must be updated during every write, which restricts random performance.

## 4. DETAILED DECONSTRUCTION OF RAID LEVELS: RAID 5, RAID 6, & RAID 10

### 4.1 RAID 5: Block-Level Striping with Distributed Parity

RAID 5 is the standard choice for enterprise datacenters. It uses block-level striping and distributes calculated parity blocks across all available drives in a rotating pattern, removing the dedicated parity bottleneck.

```
RAID 5 (DISTRIBUTED PARITY):
Disk 1: [ Block A1 ] [ Block B1 ] [ Block C1 ] [ Parity DP ]
Disk 2: [ Block A2 ] [ Block B2 ] [ Parity CP ] [ Block D1 ]
Disk 3: [ Block A3 ] [ Parity BP ] [ Block C2 ] [ Block D2 ]
Disk 4: [ Parity AP ] [ Block B3 ] [ Block C3 ] [ Block D3 ]
```

- **Working Principle:** Parity is interleaved dynamically across all drives. A single drive failure is tolerated without data loss. Rebuilding occurs by running XOR across the surviving drives.

- **Advantages & Limitations:** Highly balanced cost, performance, and protection. Storage efficiency is high:  $(N-1)/N$ . However, write operations suffer from a write penalty, and rebuilding degraded arrays can be slow.

### 4.2 RAID 6: Block-Level Striping with Dual Distributed Parity

RAID 6 extends the protection of RAID 5 by calculating two distinct parity blocks (P and Q) and distributing them across all physical disks, allowing the system to tolerate two simultaneous drive failures.

```
RAID 6 (DUAL DISTRIBUTED PARITY):
Disk 1: [ A1 ] [ B1 ] [ C1 ] [ DP ]      Disk 4: [ AP ] [ BQ ] [ C2 ]
Disk 2: [ A2 ] [ B2 ] [ CP ] [ D1 ]      Disk 5: [ AQ ] [ B3 ] [ C3 ]
Disk 3: [ A3 ] [ BP ] [ CQ ] [ D2 ]      (Tolerates two simultaneous failures)
```

- **Working Principle:** Dual mathematical algorithms (e.g., Reed-Solomon codes) generate independent parities written to separate disks.

- **Advantages & Limitations:** Superior reliability, making it ideal for massive capacity storage pools. However, it suffers from a larger write penalty (6 disk I/O operations) and slow rebuilding times.

### 4.3 RAID 10: Nested Stripe of Mirrors (RAID 1+0)

RAID 10 combines the performance of striping (RAID 0) with the redundancy of mirroring (RAID 1), creating a Stripe of Mirrors.

```
RAID 10 (STRIPED MIRRORS):
RAID 1 Mirror Group A (D1 & D2): [ Block A1 ] [ Block A3 ]
RAID 1 Mirror Group B (D3 & D4): [ Block A2 ] [ Block A4 ]
RAID 0 Stripe (Global Across Groups A & B)
```

- **Working Principle:** Drives are paired and mirrored, and those mirrored sets are striped globally. Rebuilds are fast because they only require copying data from the surviving mirrored partner.

- **Advantages & Limitations:** Extreme read/write speeds with no parity calculation latency, making it ideal for high-transaction workloads. However, it requires a minimum of 4 drives and has a 50% storage overhead.

## 5. COMPARATIVE ANALYSIS, WRITE PENALTY, & SELECTION GUIDELINES

| RAID    | Min Disks | Fault Tolerance | Write Penalty | Efficiency | Best Workloads                            |
|---------|-----------|-----------------|---------------|------------|-------------------------------------------|
| RAID 0  | 2         | 0 (None)        | 1 (None)      | 100%       | Temporary workspace, scratch datasets, i  |
| RAID 1  | 2         | 1 Drive         | 2             | 50%        | Operating systems, logging, metadata ind  |
| RAID 3  | 3         | 1 Drive         | 4             | (N-1)/N    | Large sequential streaming, video playbac |
| RAID 5  | 3         | 1 Drive         | 4             | (N-1)/N    | Database reads, general virtual machines  |
| RAID 6  | 4         | 2 Drives        | 6             | (N-2)/N    | Large capacity storage arrays, high-densi |
| RAID 10 | 4         | Up to 2 Drives  | 2             | 50%        | High-transaction databases, online transa |

### 5.1 RAID Performance Profile & Storage Efficiency Visualisation

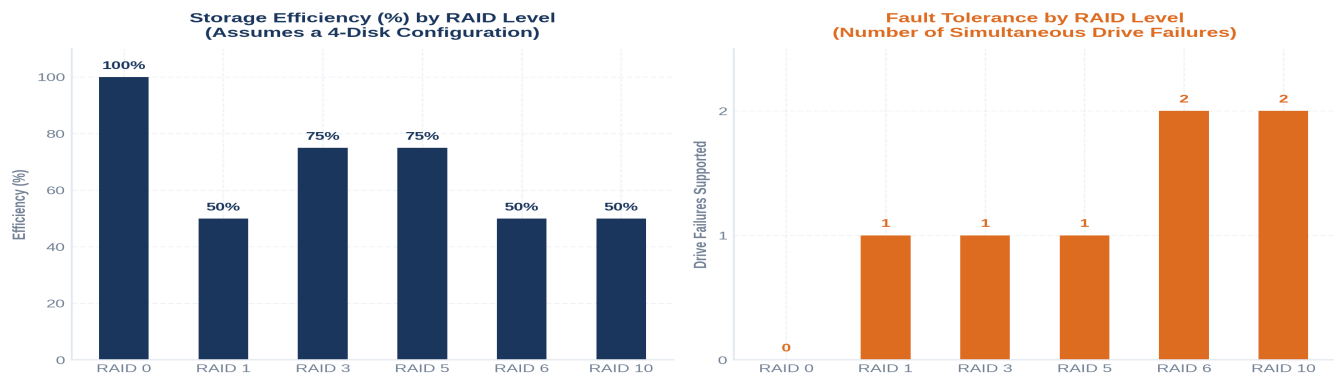


Figure 5.1: Comparative Storage Efficiency and Fault Tolerance thresholds across standard RAID levels (4-disk model where applicable).

### 5.2 Deconstructing the Parity RAID 'Write Penalty'

In parity-based arrays (RAID 5 and RAID 6), updating data blocks requires updating corresponding parity blocks, which introduces a mathematical 'Write Penalty'. **RAID 5 Write Penalty (4 I/Os):** Requires a Read-Modify-Write cycle consisting of 4 disk operations: (1) Read old data, (2) Read old parity, (3) Calculate new parity using XOR, (4) Write new data, and (5) Write new parity. **RAID 6 Write Penalty (6 I/Os):** Requires 6 disk operations due to dual independent parities: (1) Read old data, (2) Read old P parity, (3) Read old Q parity, (4) Calculate new parities, (5) Write new data, (6) Write new P, and (7) Write new Q. This penalty heavily impacts random write performance in parity systems.

### 5.3 Enterprise Architectural Selection Guidelines

For high-availability enterprise environments, the choice of RAID must align with workload patterns. For databases with high-frequency, small-block transactions (OLTP), **RAID 10** is highly recommended due to its low write penalty (2 I/Os) and lack of parity calculation latency. For standard virtual machines and read-intensive filesystems, **RAID 5** provides the ideal balance between capacity savings and reliable performance. When large-capacity SATA drives are deployed, **RAID 6** is required to protect against dual-disk failures during lengthy rebuilding cycles.